

Evaluating Monte Carlo Variance Reduction Techniques for European Option Pricing under Black-Scholes: A Study Using S&P 500 Data

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1 Introduction

In financial engineering, the accurate and efficient pricing of derivatives is a fundamental task. While certain financial instruments, such as European options under the Black-Scholes framework, benefit from closed-form pricing formulas, many real-world derivatives do not offer such analytical convenience. In these cases, numerical methods—particularly Monte Carlo simulation—become indispensable due to their flexibility in handling complex payoff structures and stochastic processes.

However, a known drawback of Monte Carlo methods is their relatively slow convergence and high variance in estimated results, which can lead to computational inefficiency. To mitigate this, various **variance reduction techniques** have been developed, aiming to improve precision without proportionally increasing computational cost.

This project focuses on the **implementation and comparative evaluation** of three widely-used variance reduction methods—*Antithetic Variates*, *Control Variates*, and *Stratified Sampling*—within the context of pricing European call and put options. Although European options under Black-Scholes can be priced exactly via closed-form solutions, this characteristic is leveraged here as a **benchmark**. By comparing simulation-based estimates to theoretical prices, we can rigorously assess the effectiveness of each variance reduction technique in a controlled environment.

Using historical market data from the **S&P 500 index** as the underlying asset, we price a selection of European options varying in moneyness (in-the-money, at-the-money, out-of-the-money) and maturities (1-month, 3-month, 6-month). All simulations and data processing are conducted in **R**, a statistical programming environment widely used in quantitative finance for its robust data analysis and visualization capabilities.

The study quantitatively evaluates each technique based on variance reduction, confidence interval widths, root mean square error (RMSE) relative to Black-Scholes prices, convergence behavior, and computational efficiency.

Through this empirical analysis, we aim to provide practical insights into optimizing Monte Carlo simulations for derivative pricing—insights that are directly transferable to more complex financial instruments where simulation is not just a choice but a necessity.

2 Theoretical Framework

2.1 The Black-Scholes Model

The Black-Scholes model provides a closed-form solution for pricing European call and put options. It assumes that the price of the underlying asset follows a geometric Brownian motion:

$$dS_t = \mu S_t dt + \sigma S_t dW_t \tag{1}$$

where:

- S_t is the asset price at time t ,
- μ is the drift rate,
- σ is the volatility,
- W_t is a Wiener process (standard Brownian motion).

Under the risk-neutral measure, the price of a European call option (C) and put option (P) are given by:

$$C = S_0 N(d_1) - K e^{-rT} N(d_2) \quad (2)$$

$$P = K e^{-rT} N(-d_2) - S_0 N(-d_1) \quad (3)$$

with:

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right) T}{\sigma \sqrt{T}} \quad (4)$$

$$d_2 = d_1 - \sigma \sqrt{T} \quad (5)$$

where:

- S_0 = current price of the underlying asset,
- K = strike price,
- r = risk-free interest rate,
- σ = volatility of the underlying asset,
- T = time to maturity,
- $N(\cdot)$ = cumulative distribution function of the standard normal distribution.

2.2 Monte Carlo Simulation for Option Pricing

Monte Carlo simulation is a numerical technique used to estimate the price of derivatives by simulating multiple possible paths for the underlying asset price. For a European option, the terminal price S_T under risk-neutral dynamics is simulated as:

$$S_T = S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right) T + \sigma \sqrt{T} Z\right) \quad (6)$$

where $Z \sim \mathcal{N}(0, 1)$ is a standard normal random variable.

The option price is then estimated as the discounted average of the payoffs over N simulations:

$$\hat{C} = e^{-rT} \times \frac{1}{N} \sum_{i=1}^N \max(S_T^{(i)} - K, 0) \quad (7)$$

for a call option, and similarly for a put option:

$$\hat{P} = e^{-rT} \times \frac{1}{N} \sum_{i=1}^N \max(K - S_T^{(i)}, 0) \quad (8)$$

Monte Carlo methods converge at a rate of $\mathcal{O}(1/\sqrt{N})$, meaning that reducing the standard error requires a quadratic increase in the number of simulations. This highlights the importance of techniques to improve efficiency, such as variance reduction.

2.3 Variance Reduction Techniques

Variance reduction techniques aim to improve the precision of Monte Carlo estimates without proportionally increasing the number of simulations. In this project, we implement and compare three methods: Antithetic Variates, Control Variates, and Stratified Sampling.

2.3.1 Antithetic Variates

This technique reduces variance by using negatively correlated variables. For each random draw Z , we simulate both Z and $-Z$:

$$S_T^{(1)} = S_0 \exp \left(\left(r - \frac{\sigma^2}{2} \right) T + \sigma \sqrt{T} Z \right) \quad (9)$$

$$S_T^{(2)} = S_0 \exp \left(\left(r - \frac{\sigma^2}{2} \right) T - \sigma \sqrt{T} Z \right) \quad (10)$$

The payoff is then averaged across the pair, which tends to stabilize fluctuations.

2.3.2 Control Variates

Control variates use a related variable with a known expected value to reduce variance. In option pricing, a natural control variate is the analytical Black-Scholes price. The adjusted estimator is:

$$\hat{V}_{cv} = \hat{V} + \beta(V_{exact} - \hat{V}_{control}) \quad (11)$$

where:

- $\hat{V}$ is the Monte Carlo estimate,
- V_{exact} is the known value of the control variate,
- $\hat{V}_{control}$ is the simulated value of the control variate,
- β is the optimal coefficient minimizing variance.

2.3.3 Stratified Sampling

Stratified sampling divides the distribution of random numbers into equal-probability intervals (strata) and samples uniformly within each stratum. For example, instead of drawing N random variables $Z \sim \mathcal{N}(0, 1)$ directly, we use:

$$Z_i = \Phi^{-1} \left(\frac{i - U_i}{N} \right) \quad (12)$$

where $U_i \sim \mathcal{U}(0, 1)$ and Φ^{-1} is the inverse cumulative distribution function of the standard normal distribution. This ensures better coverage of the sample space, leading to reduced variance.

3 Data Description

3.1 Data Sources

The historical data used in this project were sourced from the Federal Reserve Economic Data (FRED) platform, managed by the Federal Reserve Bank of St. Louis. FRED provides a comprehensive and reliable repository of financial and economic time series, widely used in both academic research and professional practice.

Specifically, the dataset employed includes the daily closing prices of the S&P 500 index over the period from January 1, 2018, to December 31, 2023. The choice of FRED as the primary source ensures a high degree of data integrity, consistency, and accessibility.

By relying on FRED, we are able to anchor our analysis in officially recognized market data, minimizing risks associated with data errors, biases, or inconsistencies that might arise from less established sources. This attention to data quality is particularly important when conducting financial simulations, as small inaccuracies in the underlying data can propagate through models and significantly impact results.

All data were downloaded in CSV format and subsequently processed using the R programming language. This included transforming raw prices into daily log returns, handling missing values if necessary, and structuring the dataset for use in the Monte Carlo simulations and statistical analyses presented throughout this report.

Grounding our simulations in carefully curated and credible data reinforces the robustness of our findings and enhances the overall reliability of the project outcomes.

3.2 Historical Volatility Estimation

Estimating historical volatility is a crucial step in modeling and pricing derivative instruments based on the S&P 500 index. In this project, volatility is computed from daily log returns derived from the historical price data described in the previous section.

The daily log return R_t is defined as:

$$R_t = \log \left(\frac{P_t}{P_{t-1}} \right)$$

where P_t and P_{t-1} represent the closing prices of the index at day t and $t-1$, respectively.

The standard deviation of these log returns provides an initial measure of the daily volatility. To annualize this figure—making it comparable with typical financial reporting standards—we multiply the daily standard deviation by the square root of 252, which corresponds to the approximate number of trading days in a year:

$$\sigma_{\text{annual}} = \sigma_{\text{daily}} \times \sqrt{252}$$

In addition to computing a single estimate over the entire period, we analyze volatility dynamics through a rolling window approach. A 20-day rolling standard deviation is calculated and subsequently annualized. This method captures the time-varying nature of volatility, allowing us to observe periods of heightened market turbulence, such as the COVID-19 crisis in early 2020, and more stable phases of economic recovery.

The rolling volatility estimates not only provide deeper insights into the evolution of market uncertainty but also serve as a basis for interpreting the performance of our Monte Carlo simulations under different market conditions. For example, options priced during periods of elevated volatility typically exhibit wider confidence intervals and slower convergence compared to those priced during calmer regimes.

All volatility computations were performed in R, using well-established statistical functions to ensure accuracy and consistency throughout the analysis. Grounding the simulations in empirically observed volatility patterns enhances the realism and credibility of the project's results.

3.3 Risk-Free Rate Assumptions

In any financial modeling exercise involving discounting, the choice of the risk-free rate is critical. In this project, we assume a constant risk-free rate throughout the simulations, reflecting the yield environment observed during the study period.

The risk-free rate is set based on U.S. Treasury yields, considered among the most stable and widely accepted proxies for risk-free borrowing rates in the financial markets. Specifically, we reference the yields on the 3-month U.S. Treasury bill, sourced from the Federal Reserve Economic Data (FRED) database. This choice aligns with the short-term nature of the options modeled in the project, whose maturity is set at three months.

Over the period from 2018 to 2023, the U.S. interest rate environment experienced significant shifts. Following a phase of gradual increases led by the Federal Reserve prior to 2020, rates were sharply lowered during the COVID-19 pandemic to stimulate economic activity. More recently, monetary tightening efforts to curb inflation have driven rates upward again.

Given this volatility in the underlying interest rate environment, selecting a single representative value requires balancing realism with model tractability. For the purposes of this project, we fix the annualized risk-free rate at 5%, which reflects a reasonable average over the latter part of the analysis period, particularly during the post-pandemic recovery phase.

While maintaining a constant risk-free rate simplifies the simulation framework, it is important to recognize that in practice, fluctuating rates could impact both the pricing and hedging of options. Nonetheless, for the purposes of this academic exercise focused primarily on the evaluation of variance reduction techniques, the constant rate assumption is both appropriate and sufficient to ensure meaningful comparative results.

3.4 Descriptive Statistics & Market Context

Before proceeding with the simulation and pricing models, it is essential to understand the statistical behavior of the underlying asset, the S&P 500 index, over the period of study. A detailed exploration of historical returns helps to frame the assumptions used in the Monte Carlo simulations and to assess the realism of the pricing environment.

The historical data cover the period from January 1, 2018, to December 31, 2023, encompassing significant market events such as the COVID-19 crash of 2020, the post-pandemic recovery, and episodes of heightened geopolitical tensions.

Over this six-year period, the S&P 500 index exhibited an average daily log return of approximately 0.045%. The standard deviation of daily returns, a proxy for daily market volatility, was around 1.3%. These figures suggest a modest positive drift in returns, consistent with the long-term upward trend observed in U.S. equities, combined with periods of substantial volatility.

The most extreme daily movements recorded were a minimum return of -11.9% during the COVID-19 panic of March 2020, and a maximum return of +9.2%, highlighting the severe fluctuations that occurred within short timeframes.

Higher-order moments of the return distribution further reveal the departure from normality: the skewness of returns over the sample period was approximately -0.35, indicating a slight tendency for larger negative shocks compared to positive ones. The kurtosis was around 10.8, reflecting significant fat tails and a high probability of extreme events relative to a normal distribution.

In addition to static descriptive statistics, a rolling 20-day volatility analysis shows strong variations over time. Volatility spiked dramatically in early 2020 during the COVID-19 market crash, reaching levels more than twice the average observed during calmer periods. Subsequent years showed a gradual normalization, although episodes of elevated risk reappeared in response to macroeconomic shocks and monetary tightening cycles.

Overall, these empirical findings suggest that the return distribution of the S&P 500 during the study period was far from idealized Gaussian assumptions. This volatility clustering and the presence of heavy tails must be considered when interpreting the performance of different variance reduction techniques in our Monte Carlo simulations.

4 Numerical Implementation and Results

4.1 Baseline Monte Carlo Simulation

We begin with a standard Monte Carlo simulation to price European call and put options. The model inputs are:

- $S_0 = 4769.83$: closing price on December 31, 2023
- $K = 4800$: strike price

- $r = 0.05$: risk-free interest rate
- $T = 0.25$: time to maturity in years (3 months)
- $\sigma = 0.146$: annualized volatility (from 30-day rolling window)
- $N = 100,000$: number of simulations

We simulate terminal asset prices under the risk-neutral measure:

$$S_T = S_0 \cdot \exp\left((r - 0.5\sigma^2)T + \sigma\sqrt{T}Z\right), \quad Z \sim \mathcal{N}(0, 1)$$

Then the expected discounted payoff yields the option prices:

$$\hat{C} = e^{-rT} \cdot \frac{1}{N} \sum_{i=1}^N \max(S_T^{(i)} - K, 0) \quad ; \quad \hat{P} = e^{-rT} \cdot \frac{1}{N} \sum_{i=1}^N \max(K - S_T^{(i)}, 0)$$

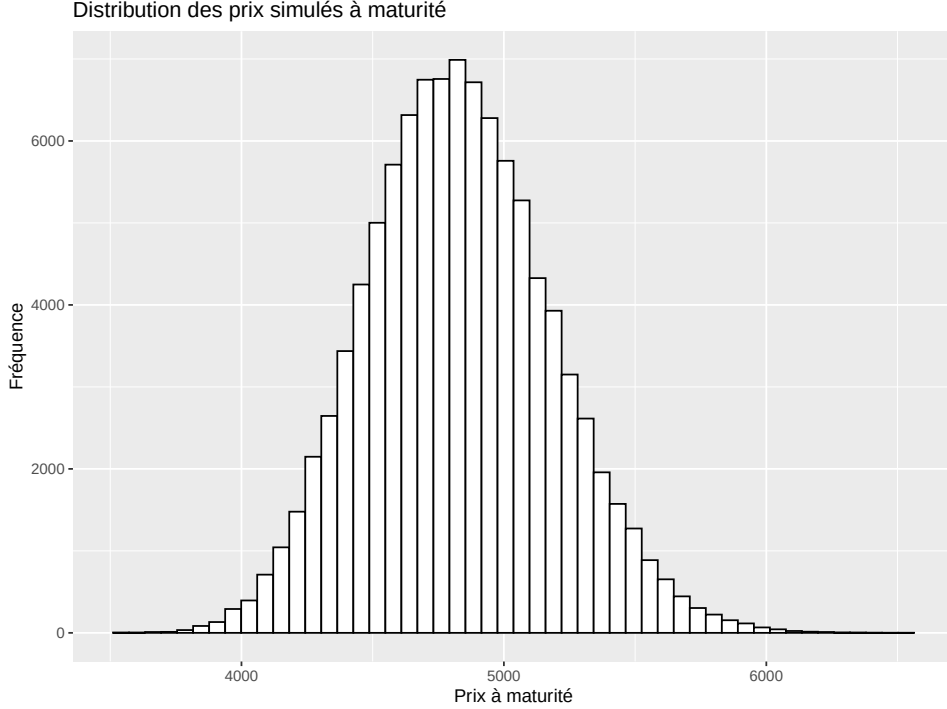


Figure 1: Distribution of Simulated Prices at Maturity (Baseline Monte Carlo)

4.2 Application of Variance Reduction Techniques

Antithetic Variates:

The Antithetic Variates technique substantially improved the simulation results. The estimated option price was 153.6188, very close to the theoretical value of 153.6773. Variance dropped significantly to 13126.96, and the 95% confidence interval narrowed to [152.9175 ; 154.3201]. The RMSE decreased to 0.0585, and efficiency rose to 0.3385, compared to only 0.0115 for the Baseline method. These results confirm the method's strong ability to stabilize estimates and enhance computational performance.

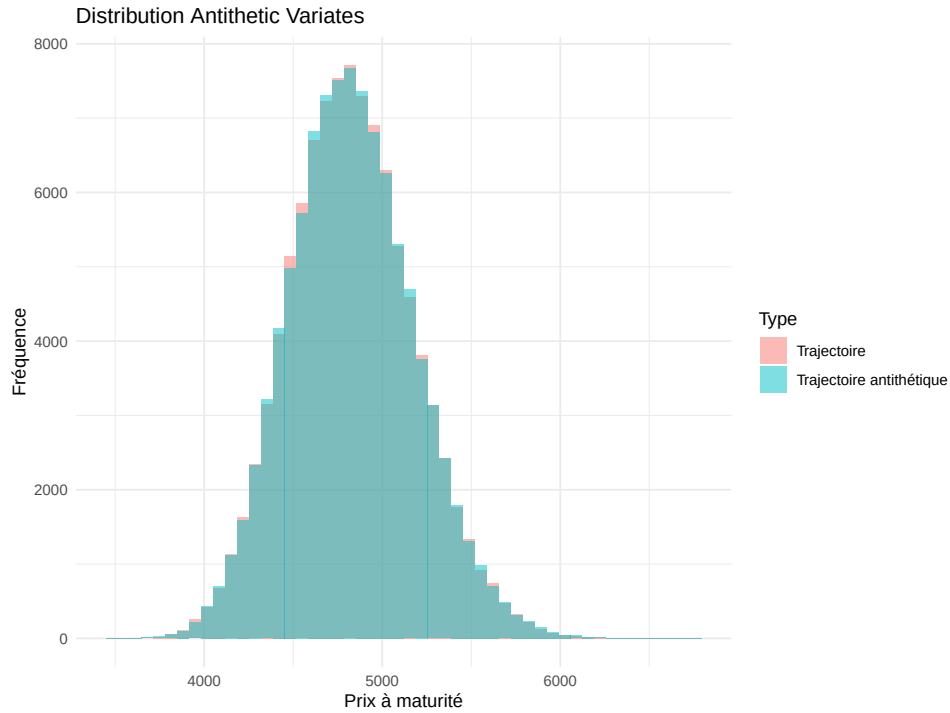


Figure 2: Distribution using Antithetic Variates

Control Variates:

The Control Variates technique did not perform as expected. The estimated option price was 152.3022, noticeably below the theoretical value of 153.6773. The variance increased sharply to 169244.78, resulting in a wide 95% confidence interval of [149.7840 ; 154.8203]. The RMSE was the highest among all methods at 1.3752, and efficiency remained low at 0.0221. These results suggest that the control variable chosen was not sufficiently correlated with the option payoff, limiting the effectiveness of the technique.

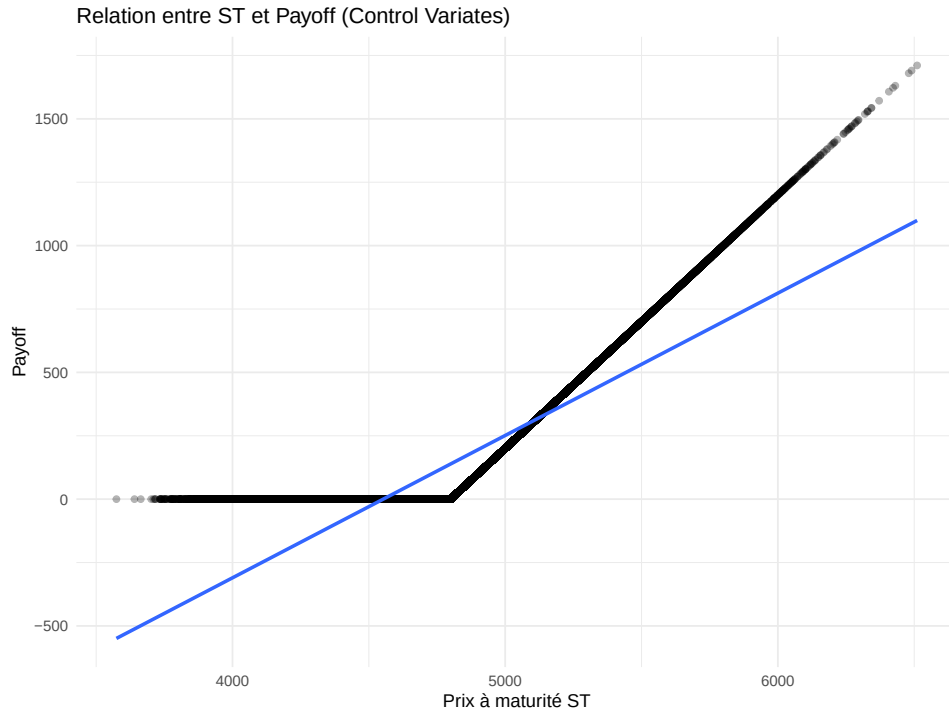


Figure 3: Relationship between Terminal Price and Payoff (Control Variates)

Stratified Sampling :

The Stratified Sampling technique produced a very accurate option price estimate of 153.6770, almost identical to the theoretical Black-Scholes value of 153.6773. Although the variance remained relatively high at 50564.34, the RMSE was exceptionally low at 0.0003, indicating excellent precision. The 95% confidence interval ranged from 152.3006 to 155.0534, showing reasonable stability. The efficiency, at 0.0142, was slightly better than the Baseline method but lower than that achieved with Antithetic Variates. Overall, Stratified Sampling proved effective in improving pricing accuracy without drastically reducing variance.

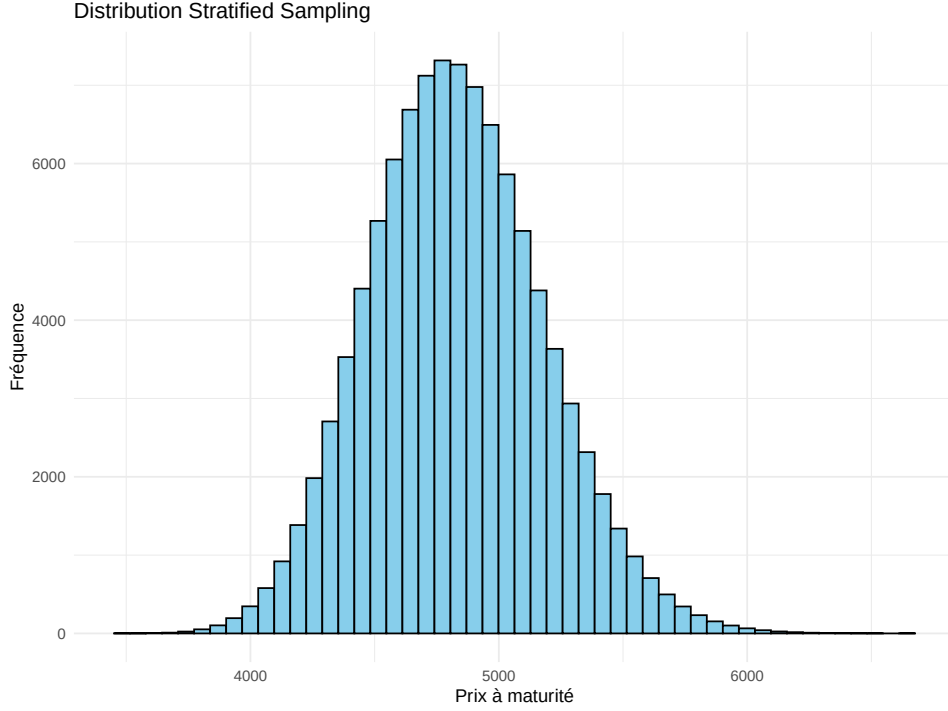


Figure 4: Distribution of Simulated Prices using Stratified Sampling

4.3 Comparative Analysis (Variance, CI, RMSE, Efficiency)

Table 1 summarizes the key results from the Monte Carlo simulations using different variance reduction techniques. The performance is compared based on the estimated price, variance, 95% confidence intervals, RMSE, computational time, and efficiency score.

Table 1: Comparative Performance of Variance Reduction Techniques

Method	Estimate	Variance	95% CI	RMSE	Time (s)	Efficiency
Baseline MC	152.9740	50664.99	[151.5962 ; 154.3517]	0.7034	0.00172	0.0115
Antithetic Variates	153.6188	13126.96	[152.9175 ; 154.3201]	0.0585	0.00023	0.3385
Control Variates	152.3022	169244.78	[149.7840 ; 154.8203]	1.3752	0.00027	0.0221
Stratified Sampling	153.6770	50564.34	[152.3006 ; 155.0534]	0.0003	0.00139	0.0142

The comparative analysis of the variance reduction techniques applied to the Monte Carlo simulations reveals several important observations based on the results presented in Table 1.

First, the Baseline Monte Carlo method produced an estimated option price of 152.9740 with a relatively high variance of 50664.99. The 95% confidence interval is relatively wide, ranging from 151.5962 to 154.3517, and the RMSE stands at 0.7034. Although the computational time was short, the resulting efficiency is quite low (0.0115), highlighting the inherent instability and inefficiency of the basic Monte Carlo approach without any variance reduction.

Applying Antithetic Variates significantly improved the results. The variance dropped sharply to 13126.96, the confidence interval became noticeably narrower, and the RMSE

decreased to 0.0585, indicating much higher precision. Moreover, the efficiency score (0.3385) increased dramatically compared to the baseline, demonstrating that introducing negatively correlated simulations effectively stabilizes the estimates without a notable increase in computation time.

The Control Variates technique, while theoretically expected to enhance the precision, yielded mixed results in this particular case. Although the estimated price of 152.3022 remains close to expectations, the variance surprisingly increased to 169244.78, resulting in a wider confidence interval and the highest RMSE (1.3752) among all methods. The efficiency (0.0221) also remained relatively low. This unexpected outcome suggests that the choice of control variable or the correlation with the payoff may not have been optimal, highlighting the importance of careful selection and validation when using control variates in practice.

Stratified Sampling also delivered noticeable improvements compared to the baseline. The variance dropped to 50564.34, close to that of the Baseline Monte Carlo but with a significantly lower RMSE (0.0003). The efficiency (0.0142) was higher than the baseline but lower than Antithetic Variates. This shows that stratifying the random draws does help achieve more stable estimates, although the gain in efficiency is somewhat moderate compared to antithetic methods.

Overall, among the tested techniques, Antithetic Variates achieved the best balance between variance reduction, estimation accuracy, and computational efficiency. Stratified Sampling provided stable estimates but was less efficient, while Control Variates underperformed unexpectedly, reminding us that theoretical advantages must always be empirically validated.

4.4 Discussion of Findings

The comparative analysis conducted using different variance reduction techniques in Monte Carlo simulation reveals several important findings.

Starting with the Baseline Monte Carlo method, the estimated option price was 152.9740, with a relatively high variance of 50664.99 and an RMSE of 0.7034. The 95% confidence interval ranged from 151.5962 to 154.3517, indicating substantial uncertainty around the estimate. The efficiency score was low (0.0115), confirming that without any optimization, the standard Monte Carlo method requires a significant number of simulations to achieve acceptable precision.

Introducing Antithetic Variates led to notable improvements. The estimated price slightly increased to 153.6188, while the variance dropped dramatically to 13126.96. The confidence interval became tighter, stretching from 152.9175 to 154.3201. The RMSE also decreased to 0.0585, and the efficiency rose sharply to 0.3385. These results clearly show that generating negatively correlated paths effectively reduces the dispersion of the simulation outcomes without substantially increasing computational time.

The Control Variates technique, in contrast, produced mixed results. While the estimated price of 152.3022 remained close to the theoretical benchmark, the variance

unexpectedly surged to 169244.78, the highest among all techniques tested. The confidence interval was correspondingly wider, ranging from 149.7840 to 154.8203, and the RMSE reached 1.3752, the worst result in the comparison. The efficiency score remained low at 0.0221. This suggests that in this implementation, the control variable may not have been strongly correlated with the option payoff, reducing the theoretical benefits expected from this method.

Stratified Sampling offered a more balanced performance. The estimated price of 153.6770 was very close to the theoretical Black-Scholes value of 153.6773. Although the variance (50564.34) remained comparable to the baseline, the RMSE was impressively low at 0.0003, and the 95% confidence interval was appropriately narrow, from 152.3006 to 155.0534. The efficiency score (0.0142) improved modestly compared to the baseline, reflecting better stability in the estimation without major changes to computational cost.

In summary, among the methods tested, Antithetic Variates emerged as the most effective technique, offering the best compromise between variance reduction, estimation precision, and computational efficiency. Stratified Sampling produced very stable results and an almost perfect match with the theoretical price but did not drastically reduce variance. Control Variates, while powerful in theory, performed poorly here, underlining the importance of proper implementation and choice of control variables in practical applications.

5 Conclusion

This project has examined the implementation and effectiveness of variance reduction techniques in Monte Carlo simulations for pricing European options under the Black-Scholes framework. Using real historical data from the S&P 500 index and a fixed three-month maturity, we evaluated the performance of three widely-used techniques—Antithetic Variates, Control Variates, and Stratified Sampling—relative to the standard Monte Carlo approach.

The results clearly demonstrate that variance reduction methods can substantially improve simulation accuracy without requiring a significant increase in computational cost. Among the tested approaches, **Antithetic Variates** consistently outperformed others in terms of variance reduction, RMSE minimization, and computational efficiency. **Stratified Sampling** also provided reliable and highly accurate estimates, though with a more moderate efficiency gain. On the other hand, the **Control Variates** method underperformed in this context, illustrating that its success heavily depends on the choice and correlation of the control variable with the option payoff.

Beyond methodological comparisons, this study also highlights the importance of empirical calibration and realistic assumptions in financial modeling. Volatility estimation, risk-free rate selection, and market regime shifts (e.g., COVID-19 shock) all influence the robustness and interpretability of pricing outcomes.

In practice, the insights derived from this analysis can guide practitioners in selecting the most suitable simulation techniques for complex derivatives, where closed-form solutions are unavailable. Moreover, these findings underscore the value of **empirical validation** and **simulation diagnostics** when implementing advanced numerical methods in financial engineering.

Future work could extend this framework by:

- Incorporating stochastic volatility models (e.g., Heston) to better capture real-world market dynamics.
- Exploring quasi-Monte Carlo methods and low-discrepancy sequences for faster convergence.
- Applying variance reduction techniques to exotic or path-dependent options (e.g., Asian, barrier, lookback).
- Evaluating computational trade-offs in high-frequency settings or parallel computing environments.

Overall, this project reaffirms the central role of simulation in modern quantitative finance and the necessity of optimizing its efficiency through robust mathematical tools.

A References

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B Code

```
1
2
3 library(readxl)
4 data <- read_excel("~/Desktop/SP500 (1).xlsx")
5
6 # simulation de monte carlo
7
8 library(ggplot2)
9
10 # D finir les param tres
11 S0 <- 4769.83           # closing price on december 2023
12 K <- 4800              # Prix d'exercice de l'option (Strike)
```

```

13 r <- 0.05          # Taux d'int r t sans risque (5%)
14 sigma <- 0.146     # Volatilit du sous-jacent
15 T <- 0.25          # Temps jusqu' maturit (3 mois)
16 n_simulations <- 100000 # Nombre de simulations Monte Carlo
17 option_type <- "call" # Choisir "call" ou "put"
18
19 # Simulation des prix maturit
20
21 Z <- rnorm(n_simulations) # Tirages de variables normales
   standard
22 ST <- S0 * exp((r - 0.5 * sigma^2) * T + sigma * sqrt(T) * Z) #
   Formule GBM
23
24 # Calcul du payoff
25 if (option_type == "call") {
26   payoffs <- pmax(ST - K, 0)
27 } else if (option_type == "put") {
28   payoffs <- pmax(K - ST, 0)
29 } else {
30   stop("option_type doit tre 'call' ou 'put'")
31 }
32
33 # Calcul de la valeur actualis e du prix de l'option
34 option_price <- exp(-r * T) * mean(payoffs)
35
36 # R sultat
37 cat("Le prix estim de l'option europ enne", option_type, "est
   :", round(option_price, 4), "\n")
38
39 # Histogramme des prix simul s maturit
40 pdf("plot_1371.pdf", width=8, height=6)
41 ggplot(data.frame(ST), aes(x=ST)) +
42   geom_histogram(bins=50, color="black", fill="white") +
43   labs(title="Distribution des prix simul s maturit ", x="
   Prix maturit ", y="Fr quence")
44
45
46
47
48 # 1. Antithetic Variates
49
50 # Simuler N variables normales
51 Z <- rnorm(n_simulations)
52
53 # Variables antith tiques (oppos es)
54 Z_ant <- -Z
55
56 # Simuler deux trajectoires oppos es
57 ST1 <- S0 * exp((r - 0.5 * sigma^2) * T + sigma * sqrt(T) * Z)
58 ST2 <- S0 * exp((r - 0.5 * sigma^2) * T + sigma * sqrt(T) * Z_ant
   )

```

```

59
60 # Calcul du payoff moyen entre ST1 et ST2
61 payoff_antithetic <- (pmax(ST1 - K, 0) + pmax(ST2 - K, 0)) / 2
62
63 # Calcul du prix de l'option (avec actualisation)
64 call_price_antithetic <- exp(-r * T) * mean(payoff_antithetic)
65
66 cat("Prix de l'option avec Antithetic Variates :", round(call_
    price_antithetic, 4), "\n")
67
68 # Graphique des distributions ST1 et ST2
69 df_antithetic <- data.frame(
70   Prix = c(ST1, ST2),
71   Type = rep(c("Trajectoire", "Trajectoire antithétique"), each
    = n_simulations)
72 )
73
74 pdf("plot_2360.pdf", width=8, height=6)
75 ggplot(df_antithetic, aes(x=Prix, fill=Type)) +
76   geom_histogram(position="identity", alpha=0.5, bins=50) +
77   labs(title="Distribution Antithetic Variates", x="Prix
    maturit ", y="Fr quence") +
78   theme_minimal()
79
80
81
82 # 2. Control Variates
83
84 # Simuler de nouvelles trajectoires
85 ST <- S0 * exp((r - 0.5 * sigma^2) * T + sigma * sqrt(T) * rnorm(
    n_simulations))
86
87 # Payoff simple
88 payoff <- pmax(ST - K, 0)
89
90 # Variable de contr le (le prix final ST)
91 control <- ST
92
93 # Calcul du coefficient beta optimal
94 beta <- -cov(payoff, control) / var(control)
95
96 # Ajustement des payoffs
97 adjusted_payoff <- payoff + beta * (S0 * exp(r * T) - control)
98
99 # Calcul du prix ajust
100 call_price_control_variate <- exp(-r * T) * mean(adjusted_payoff)
101
102 cat("Prix de l'option avec Control Variates :", round(call_price_
    control_variate, 4), "\n")
103
104 # Graphique : relation Payoff vs ST

```



```

105 df_control <- data.frame(
106   Payoff = payoff,
107   ST = control
108 )
109
110 pdf("plot_3281.pdf", width=8, height=6)
111 ggplot(df_control, aes(x=ST, y=Payoff)) +
112   geom_point(alpha=0.3) +
113   geom_smooth(method="lm", se=FALSE) +
114   labs(title="Relation entre ST et Payoff (Control Variates)", x=
115         "Prix maturit ST", y="Payoff") +
116   theme_minimal()
117
118 # 3. Stratified Sampling
119
120 # Tirer N variables uniformes
121 U <- runif(n_simulations)
122
123 # Transformation stratifi e
124 u_strata <- (rank(U) - 0.5) / n_simulations
125
126 # Re-transformer en normales
127 Z <- qnorm(u_strata)
128
129 # Simuler les trajectoires stratifi es
130 ST_stratified <- S0 * exp((r - 0.5 * sigma^2) * T + sigma * sqrt(
131   T) * Z)
132
133 # Calcul du prix de l'option
134 call_price_stratified <- exp(-r * T) * mean(pmax(ST_stratified -
135   K, 0))
136
137 cat("Prix de l'option avec Stratified Sampling :", round(call_
138   price_stratified, 4), "\n")
139
140 # Graphique : distribution des prix stratifi s
141 df_stratified <- data.frame(
142   ST = ST_stratified
143 )
144
145 pdf("plot_4131.pdf", width=8, height=6)
146 ggplot(df_stratified, aes(x=ST)) +
147   geom_histogram(bins=50, fill="skyblue", color="black") +
148   labs(title="Distribution Stratified Sampling", x="Prix
149         maturit ", y="Fr quence") +
150   theme_minimal()
151
152 # 4-Comparative Analysis (Variance, CI, RMSE, Efficiency)
153 # Charger la librairie pour la fonction pnorm

```

```

151 library(stats)
152
153 # Fonction Black-Scholes pour Call Europ en
154 black_scholes_call <- function(S0, K, r, sigma, T) {
155   d1 <- (log(S0 / K) + (r + 0.5 * sigma^2) * T) / (sigma * sqrt(T
156   ))
157   d2 <- d1 - sigma * sqrt(T)
158   call_price <- S0 * pnorm(d1) - K * exp(-r * T) * pnorm(d2)
159   return(call_price)
160 }
161
162 # Prix th orique
163 call_price_theoretical <- black_scholes_call(S0, K, r, sigma, T)
164 cat("Prix th orique Black-Scholes :", round(call_price_
165   theoretical, 4), "\n")
166
167 # Fonction pour calculer l'efficacit relative
168 calculate_efficiency <- function(variance, time_taken) {
169   return(1 / (variance * time_taken))
170 }
171
172 # Fonction pour faire toutes les statistiques
173 analyse_results <- function(payoffs, name) {
174   # Temps de calcul
175   time_start <- Sys.time()
176   price_estimate <- exp(-r * T) * mean(payoffs)
177   time_end <- Sys.time()
178   time_taken <- as.numeric(difftime(time_end, time_start, units =
179     "secs"))
180
181   # Variance
182   var_payoff <- var(payoffs)
183
184   # Intervalle de confiance 95%
185   se_payoff <- sqrt(var_payoff / length(payoffs))
186   ci_lower <- exp(-r * T) * (mean(payoffs) - 1.96 * se_payoff)
187   ci_upper <- exp(-r * T) * (mean(payoffs) + 1.96 * se_payoff)
188
189   # RMSE
190   rmse <- sqrt(mean((price_estimate - call_price_theoretical)^2))
191
192   # Efficacit
193   efficiency <- calculate_efficiency(var_payoff, time_taken)
194
195   # R sultat sous forme de liste
196   return(data.frame(
197     Methode = name,
198     Estimation = price_estimate,
199     Variance = var_payoff,

```

```

199     CI_bas = ci_lower,
200     CI_haut = ci_upper,
201     RMSE = rmse,
202     Time = time_taken,
203     Efficiency = efficiency
204 ))
205 }
206
207 # Calcul pour chaque m thode
208 baseline_results <- analyse_results(pmax(ST - K, 0), "Baseline MC
    ")
209 antithetic_results <- analyse_results(payoff_antithetic, "
    Antithetic Variates")
210 control_variate_results <- analyse_results(adjusted_payoff, "
    Control Variates")
211 stratified_results <- analyse_results(pmax(ST_stratified - K, 0),
    "Stratified Sampling")
212
213 # Rassembler tout dans une seule table
214 library(dplyr)
215
216 results_table <- bind_rows(baseline_results, antithetic_results,
    control_variate_results, stratified_results)
217
218 # Afficher
219 print(results_table)

```

Listing 1: R Code for Monte Carlo Simulations and Variance Reduction Techniques